

Power BI Case Study: Outbreaks in Toronto Healthcare Institutions

Operational dashboard using City of Toronto open data

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Jan-Jun 2026 Operational View

Total Outbreaks	Active	Closed	Active Rate	Avg Duration	New Last 7 Days
350	9	341	2.6%	13.2 days	1

Project Summary

This project uses City of Toronto open data to analyze outbreaks in healthcare institutions, including long-term care homes, retirement homes, and hospitals.

I built an operational Power BI dashboard to monitor active and closed outbreaks, track outbreak trends over time, compare outbreak settings and types, and identify the causative agents associated with current active outbreaks.

The dashboard was designed to answer practical questions such as how many outbreaks are active, which healthcare settings have the highest number of outbreaks, how outbreaks changed over time, and which causative agents are most common.

Project Motivation

I started this project in April 2026 because I wanted to better understand the current status of COVID-related outbreaks in Toronto healthcare institutions. COVID was no longer receiving much attention in the news, so I used public open data to explore whether outbreaks were still occurring and how the trend was changing over time.

This project gave me an opportunity to apply Power BI to a real public-health dataset and create a dashboard that supports operational monitoring, trend analysis, and clear communication of outbreak status.

Dataset and Source

The dataset used in this project is Outbreaks in Toronto Healthcare Institutions, published by the City of Toronto Open Data Portal.

Data source	City of Toronto Open Data Portal
Dataset	Outbreaks in Toronto Healthcare Institutions
Dataset link	https://open.toronto.ca/dataset/outbreaks-in-toronto-healthcare-institutions/
Open data platform	Powered by CKAN
CKAN link	https://ckan.org/

The dataset contains outbreak records from Toronto healthcare institutions, including hospitals, long-term care homes, and retirement homes. Each record represents an outbreak and includes information such as the institution name, institution address, outbreak setting, outbreak type, causative agent, outbreak start date, declared over date, and active status.

Because the data is publicly available, it is appropriate for a portfolio project and allows the analysis to be reviewed, reproduced, and extended in the future.

Business Questions

This dashboard was designed to answer practical questions about outbreak activity in Toronto healthcare institutions.

1. How many outbreaks are currently active?
2. What percentage of outbreaks are active versus closed?
3. Which healthcare settings have the highest number of outbreaks?
4. How have outbreaks changed over time?
5. Which outbreak types are most common?
6. Which causative agents are associated with active outbreaks?
7. Which active outbreaks have been open the longest?
8. How has the COVID-19 outbreak trend changed across recent years?

These questions helped guide the dashboard design, the DAX measures, and the choice of visuals. The goal was to create a report that could quickly communicate the current outbreak situation and highlight areas that may need attention.

Tools and Skills Demonstrated

This project demonstrates the use of Power BI to combine multiple public open data files into one analytical model and transform them into an operational dashboard for monitoring healthcare institution outbreaks.

Tools used

- Power BI Desktop
- Power Query
- DAX

- City of Toronto Open Data
- CKAN open data platform
- ChatGPT as an AI-assisted learning and development tool

Power BI and analytics skills demonstrated

- Combining multiple yearly CSV files into one table
- Data quality review and validation
- Data type validation for date and categorical fields
- Calendar table creation
- Interactive filtering by Month Year, status, setting, outbreak type, and causative agent
- Data modelling using a fact table and date table
- DAX measure creation
- KPI design
- Trend analysis over time
- Category breakdown by outbreak setting, outbreak type, and causative agent
- Operational dashboard design
- Visual formatting and layout planning
- Critical review of AI-generated suggestions before applying them

Dashboard Objectives

The main objective of this dashboard was to create a clear operational view of outbreaks in Toronto healthcare institutions.

The dashboard was designed to help users quickly understand:

- how many outbreaks were active
- how many outbreaks were closed
- the percentage of outbreaks still active
- how outbreaks changed over time
- which healthcare settings had the highest outbreak counts
- which outbreak types were most common
- which causative agents were associated with active outbreaks
- which active outbreaks had been open the longest

The dashboard also includes interactive filters for Month Year, status, setting, outbreak type, and causative agent. These filters allow users to focus the analysis on specific time periods, active or closed outbreaks, healthcare settings, outbreak categories, and agents.

The goal was not only to create visualizations, but to organize the information in a way that supports monitoring and quick interpretation. The dashboard combines KPI cards, trend analysis, category breakdowns, interactive filtering, and a detailed active outbreak table to move from a high-level summary to row-level details.

Data Preparation and Validation

The data was provided as yearly CSV files using the naming convention `ob_report_YYYY.csv`, where YYYY represents the year of the outbreak data.

For this project, I loaded files from 2019 onward because the analysis was focused on COVID-19 and more recent outbreak patterns. I combined the yearly files into one table to support both operational monitoring and historical trend analysis. To make the report easier to refresh on another computer, I created a Power Query parameter called `DataFolderPath`. This parameter points to the local data/raw folder, and the source step uses `Folder.Files(DataFolderPath)` instead of a hard-coded personal folder path.

The dataset quality was generally strong. The 2026 dataset was rated Gold quality, while previous years were rated Silver quality. Because of the quality of the source data, significant manual data cleaning was not required.

The main preparation steps were:

- loading multiple yearly CSV files
- filtering the files to include outbreak report CSV files
- combining the files into one outbreak table
- validating date fields such as outbreak start date and declared over date
- validating categorical fields such as setting, outbreak type, status, and causative agent
- creating a status field to support active versus closed analysis
- preparing the data model for time-based analysis using a Calendar table

One important validation step involved the causative agent field. The dataset included both COVID-19 and Coronavirus*. Although these values may appear similar at first, the data portal documentation explains that Coronavirus* refers to seasonal coronavirus and not COVID-19. For that reason, I kept the two values separate in the analysis.

This helped preserve the meaning of the original data and avoid overstating COVID-19-related outbreaks.

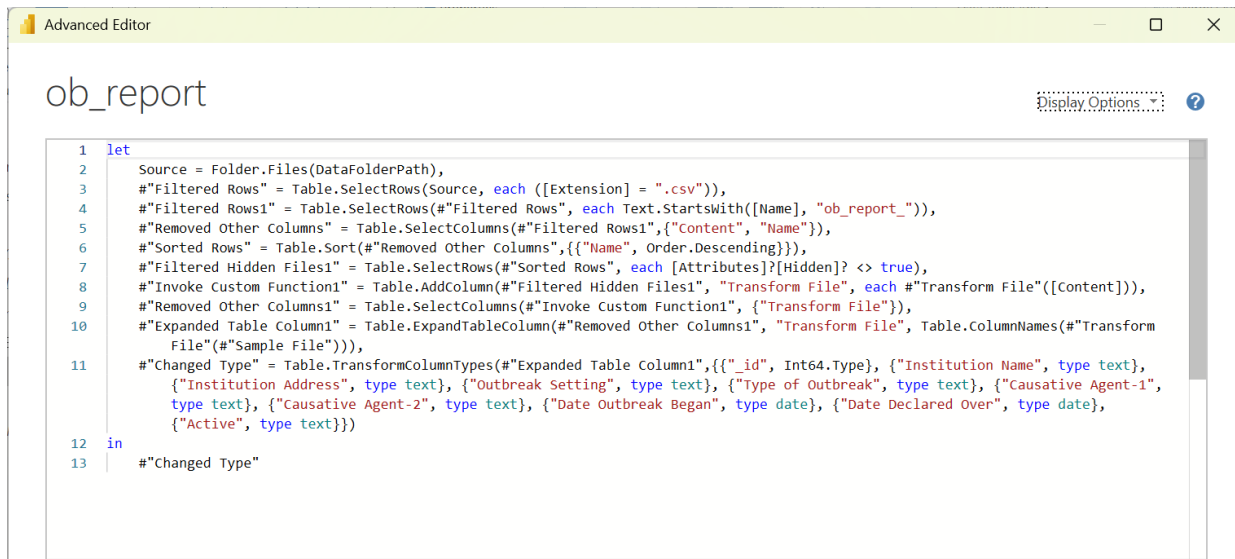


Figure 1. Power Query file combination. Yearly CSV files were loaded from a folder using the `DataFolderPath` parameter, filtered to include outbreak report files, combined into one table, and transformed by setting the correct data types.

Data Model

The dashboard uses a simple data model with one main outbreak table and a Calendar table.

The main outbreak table contains the combined records from the yearly CSV files. Each row represents one outbreak record and includes fields such as institution name, institution address, outbreak setting, outbreak type, causative agent, outbreak start date, declared over date, and active status.

A separate Calendar table was created to support time-based analysis. The Calendar table includes date attributes such as year, month, month name, month year, quarter, start of month, and week start. It is connected to the outbreak table using the outbreak start date.

This approach keeps the model clean because date attributes are handled in the Calendar table instead of being duplicated in the outbreak table. The outbreak table remains focused on event-level outbreak details, while the Calendar table supports filtering, sorting, and trend analysis over time.

This design also supports the Month Year slicer and the Outbreaks Started Over Time visual, which were important parts of the analysis.

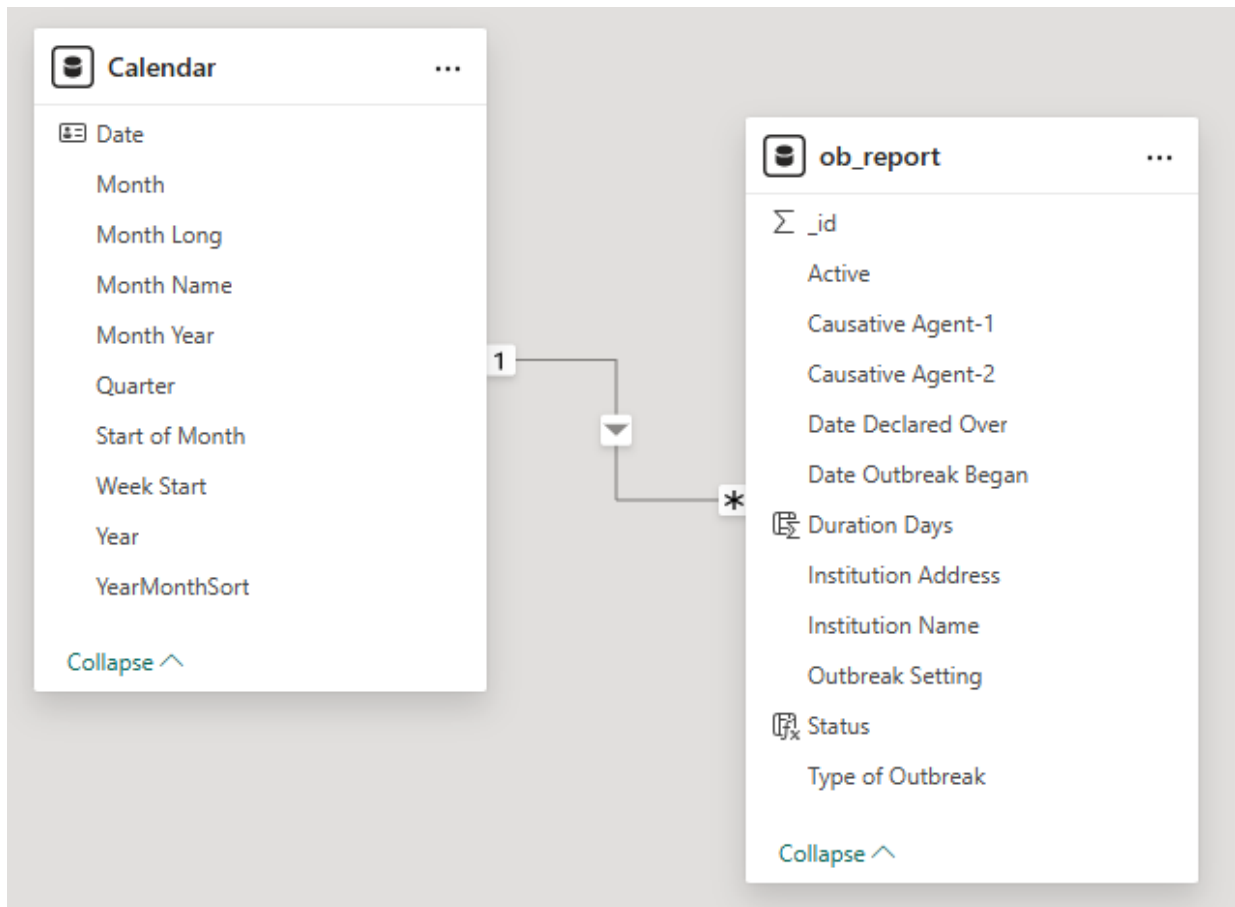


Figure 2. Data model view. The model uses a combined outbreak table connected to a Calendar table for time-based analysis. The Calendar table supports Month Year filtering, sorting, and trend analysis, while the outbreak table stores event-level outbreak records.

DAX Measures

The dashboard uses DAX measures to calculate key outbreak metrics and support interactive analysis.

The main measures created for the dashboard include:

- Total Outbreaks: counts the total number of outbreak records.
- Active Outbreaks: counts outbreaks that are currently active.
- Closed Outbreaks: counts outbreaks that are no longer active.
- Active Rate: calculates the percentage of outbreaks that are active.
- Average Duration: calculates the average outbreak duration.
- New Outbreaks Last 7 Days: counts outbreaks that started in the last seven days.
- Distinct Institutions: counts the number of unique healthcare institutions in the data.

These measures were used in KPI cards, trend charts, category breakdowns, and the active outbreak detail table.

The DAX measures make the dashboard more flexible because they respond to the report filters, including Month Year, status, setting, outbreak type, and causative agent. This allows users to analyze outbreak metrics across different time periods and categories.

A separate `_Measures` table was used to organize the DAX measures. This keeps the model easier to maintain and separates business calculations from the raw data fields.

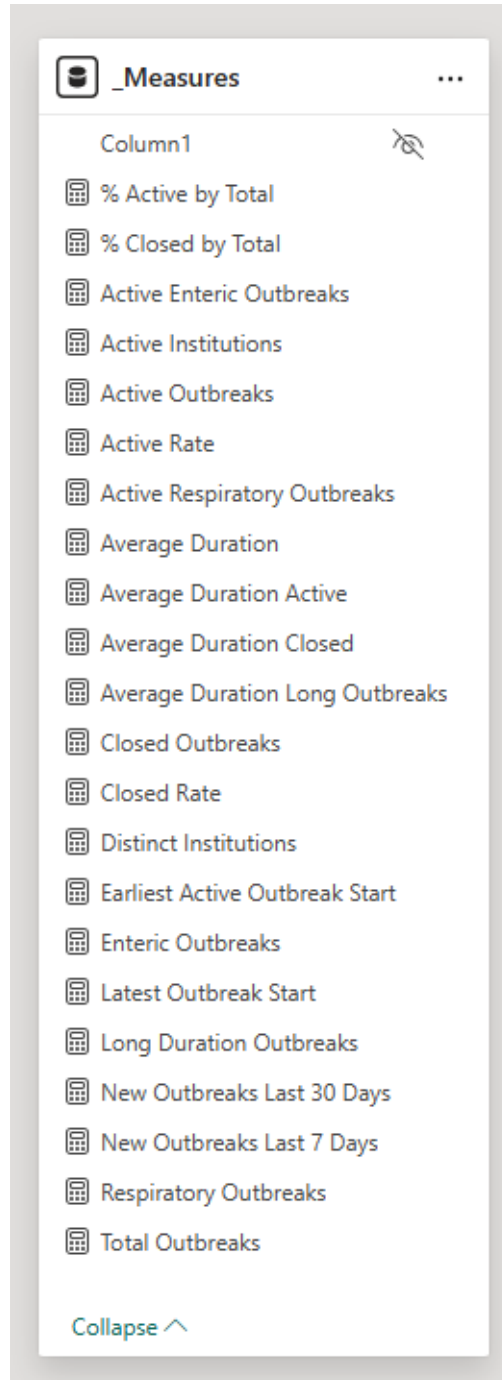


Figure 3. DAX measures table. Key measures were organized in a dedicated `_Measures` table, including total outbreaks, active outbreaks, closed outbreaks, active rate, average duration, and recent outbreak measures.

Dashboard Design

The dashboard was designed as an operational monitoring view, with the most important information placed at the top and supporting details below.

The top section includes KPI cards for key metrics such as total outbreaks, active outbreaks, closed outbreaks, active rate, average duration, and new outbreaks in the last seven days. In the January to June 2026 view, the dashboard shows 350 total outbreaks, 9 active outbreaks, 341 closed outbreaks, an active rate of 2.6%, an average duration of 13.2 days, and 1 new outbreak in the last seven days.

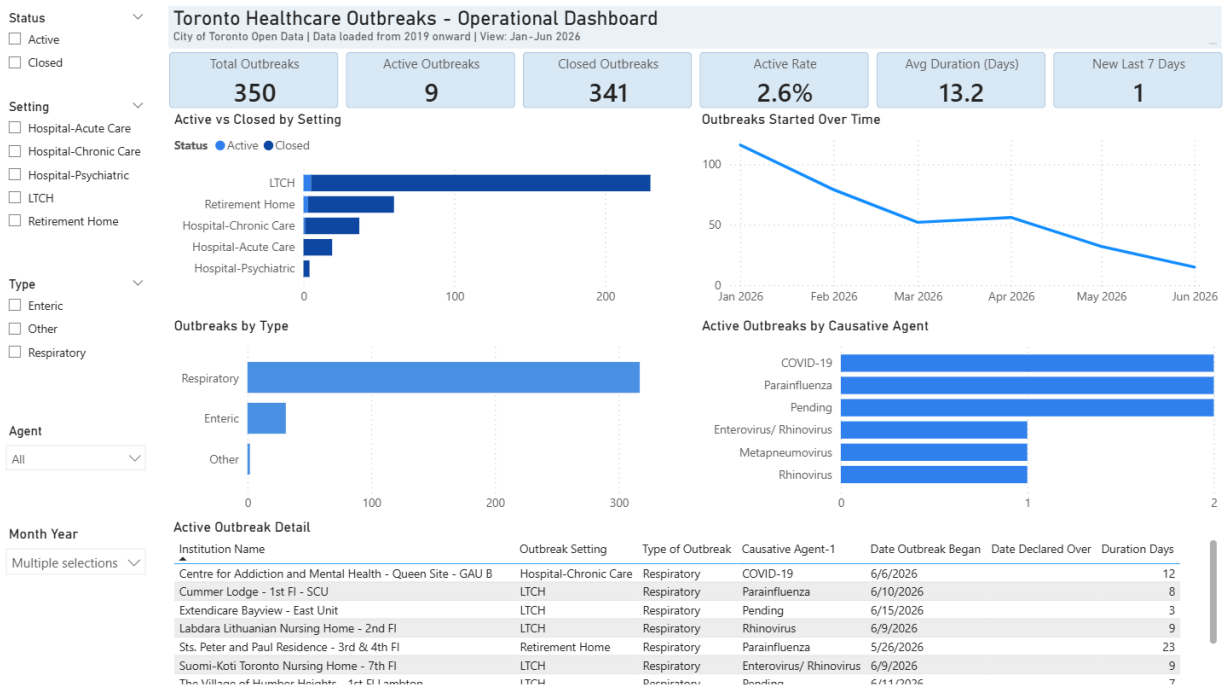


Figure 4. Dashboard overview - Jan-Jun 2026. Default operational dashboard view filtered to January to June 2026. The dashboard shows outbreak KPIs, active versus closed outbreaks by setting, monthly outbreak trend, outbreak type, active causative agents, and row-level active outbreak details.

The middle section includes visuals that help users understand patterns and distribution:

- Active vs Closed by Setting compares outbreak status across healthcare settings and shows where outbreaks are concentrated.
- Outbreaks Started Over Time shows how outbreak counts changed by month and helps identify seasonal peaks.
- Outbreaks by Type shows the distribution of outbreak categories such as respiratory, enteric, and other.
- Active Outbreaks by Causative Agent highlights which agents are associated with active outbreaks.

In the dashboard layout, Active vs Closed by Setting is placed on the left and Outbreaks Started Over Time is placed on the right. This layout shows both the current operational breakdown and the time trend side by side.

The dashboard also includes slicers for Month Year, status, setting, outbreak type, and causative agent. These filters allow users to narrow the analysis and explore specific time periods, healthcare settings, outbreak categories, or agents.

The bottom section includes an active outbreak detail table. This table provides row-level information such as institution name, setting, outbreak type, causative agent, start date, declared over date, and duration. It supports the operational purpose of the dashboard by allowing users to move from high-level metrics to specific outbreak records.

The layout was designed to guide users from summary to detail: first showing the overall situation, then the current breakdown and time trend, then category-level analysis, and finally the detailed records behind the numbers.

Key Insights

The dashboard helped identify several important patterns in the outbreak data.

One of the most useful visuals was Outbreaks Started Over Time. This visual showed a seasonal pattern, with outbreak peaks commonly occurring during the winter months, especially around December or January.

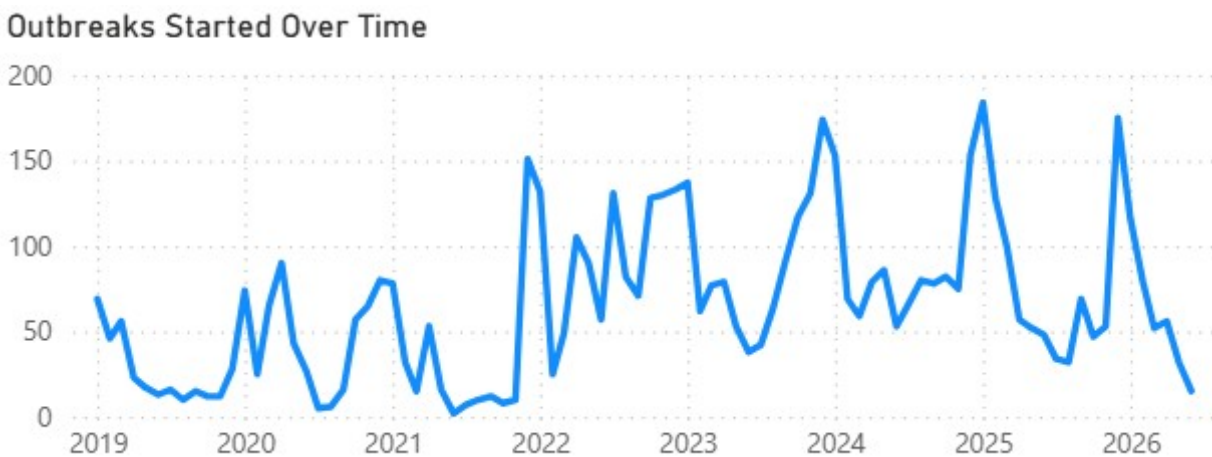


Figure 5. Outbreaks Started Over Time - 2019 onward. Historical trend showing outbreak peaks around winter months, especially December and January, across recent years.

Looking at the last three years, the total number of outbreaks showed similar winter peak patterns in 2024, 2025, and 2026. However, the 2026 trend decreased through spring and early summer, and the Jan-Jun 2026 view showed a better downward trend compared with 2025.

Outbreaks Started Over Time

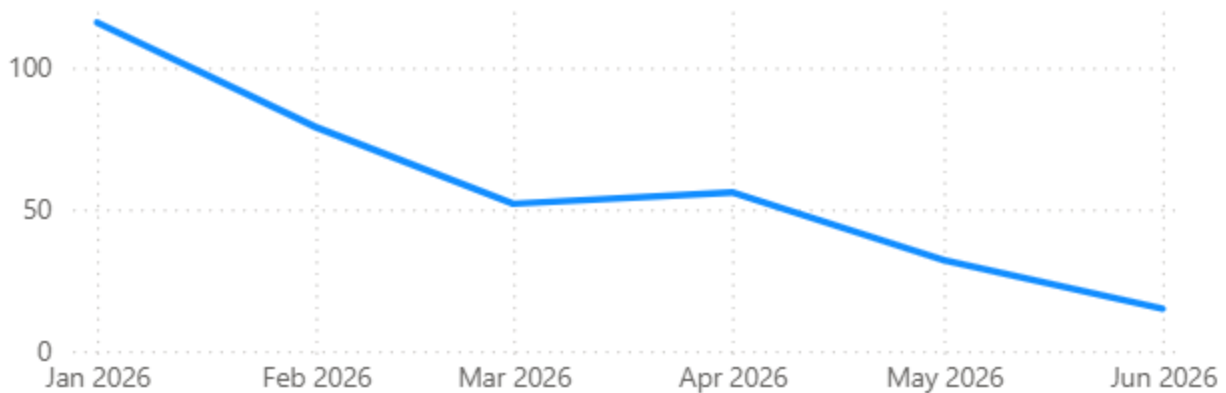


Figure 6. Outbreaks Started Over Time - Jan-Jun 2026. The 2026 view shows a downward trend after January, with lower outbreak counts through spring and early summer.

COVID-19 remained the leading causative agent in the Jan-Jun 2026 operational view. This was an important finding because COVID was receiving less attention in the news, but it was still relevant in healthcare institution outbreak monitoring.

The Agent filter was useful for analyzing COVID-19 separately. When filtering the trend to COVID-19 only, the peak number of outbreaks decreased significantly across recent winter periods: 131 in December 2023, 73 in December 2024, and 33 in January 2026.

Outbreaks Started Over Time



Figure 7. COVID-19 Outbreaks Started Over Time. Filtered trend showing COVID-19 outbreaks over time. The peak values decreased from 131 in December 2023, to 73 in December 2024, and to 33 in January 2026.

The Jan-Jun 2026 view showed that outbreak counts were going down, with June still representing a partial month at the time of analysis. This makes continued monitoring important as more 2026 data becomes available.

Additional dashboard visuals helped explain the operational view in more detail.

Active vs Closed by Setting

Status ● Active ● Closed

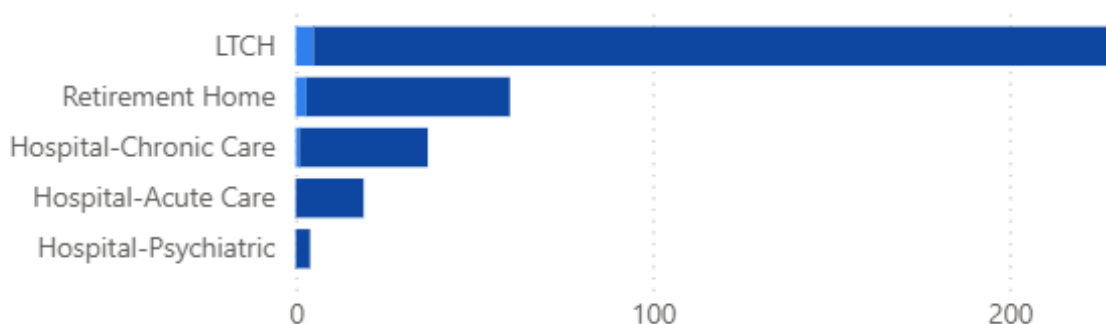


Figure 8. Active vs Closed by Setting - Jan-Jun 2026. Long-term care homes had the highest number of outbreaks in the selected period, followed by retirement homes and hospital settings.

Outbreaks by Type

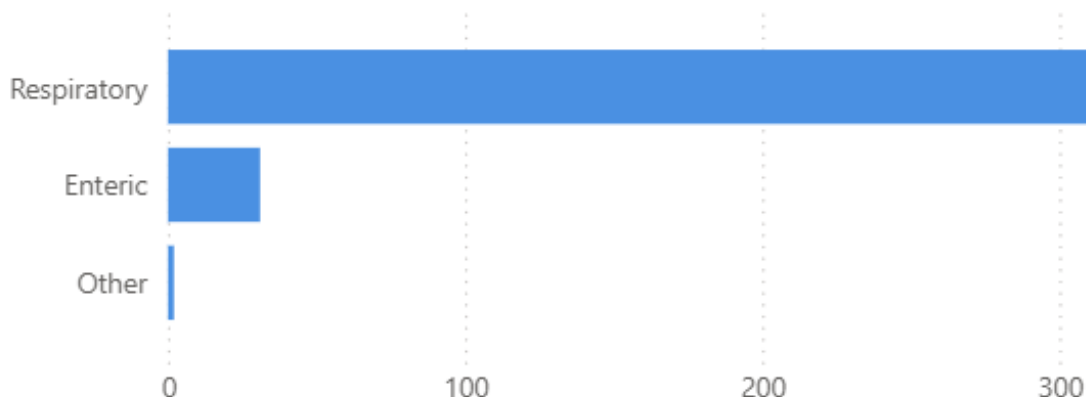


Figure 9. Outbreaks by Type - Jan-Jun 2026. Respiratory outbreaks represented the largest outbreak type in the selected period.

Active Outbreaks by Causative Agent

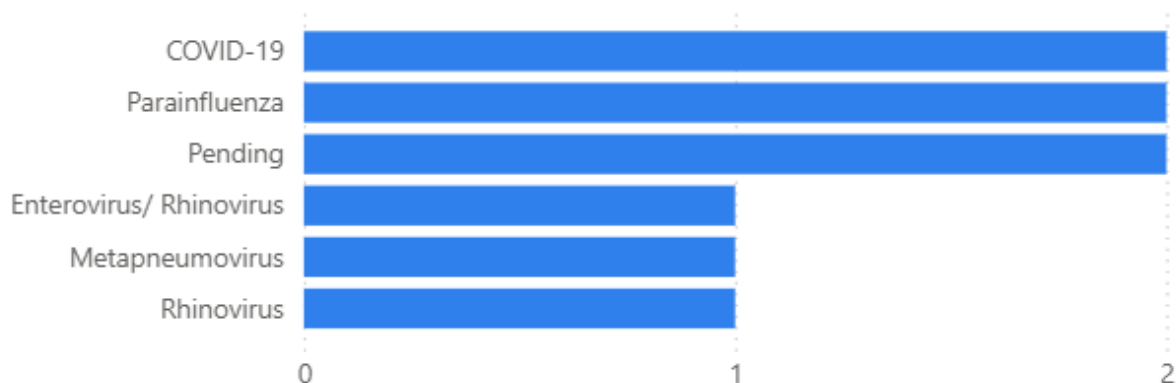


Figure 10. Active Outbreaks by Causative Agent - Jan-Jun 2026. COVID-19 was one of the leading causative agents among active outbreaks in the selected operational view.

Active Outbreak Detail

Institution Name	Outbreak Setting	Type of Outbreak	Causative Agent-1	Date Outbreak Began	Date Declared Over	Duration Days
Sts. Peter and Paul Residence - 3rd & 4th Fl	Retirement Home	Respiratory	Parainfluenza	5/26/2026		23
The Village of Humber Heights - 2nd Fl Egerton	Retirement Home	Respiratory	Metapneumovirus	6/5/2026		13
Centre for Addiction and Mental Health - Queen Site - GAU B	Hospital-Chronic Care	Respiratory	COVID-19	6/6/2026		12
Labdara Lithuanian Nursing Home - 2nd Fl	LTCH	Respiratory	Rhinovirus	6/9/2026		9
Suomi-Koti Toronto Nursing Home - 7th Fl	LTCH	Respiratory	Enterovirus/ Rhinovirus	6/9/2026		9
Cummer Lodge - 1st Fl - SCU	LTCH	Respiratory	Parainfluenza	6/10/2026		8
The Mowford - 5th Fl	Retirement Home	Respiratory	COVID-19	6/10/2026		8

Figure 11. Active Outbreak Detail - Jan-Jun 2026. Row-level detail table sorted by duration to show the longest active outbreaks first, including institution, setting, outbreak type, causative agent, start date, and duration.

Challenges and Decisions

One important decision in this project was preserving the meaning of the original dataset instead of grouping similar-looking values too quickly. For example, I kept Coronavirus* separate from COVID-19 based on the dataset documentation.

Another decision was how to structure the data model for time-based reporting. Instead of creating year and month columns directly in the outbreak table, I used a separate Calendar table. This kept the outbreak table focused on event-level data and allowed the Calendar table to handle time grouping, sorting, and Month Year filtering.

I also decided to combine multiple yearly CSV files into one table. This allowed the dashboard to support both recent operational monitoring and historical trend analysis. The operational view was especially focused on January to June 2026, while the historical data from 2019 onward helped provide context for COVID-related outbreak patterns.

A design challenge was deciding how much information to show on one page. I kept the dashboard focused on the most useful operational views: KPI cards, outbreak trends over time, active versus closed outbreaks by setting, outbreak type, causative agent, and a detail table for active outbreaks.

These decisions helped keep the dashboard clear, accurate, and useful for both quick monitoring and deeper analysis.

Responsible Use of AI

I used ChatGPT as an AI-assisted learning and development tool while building this Power BI dashboard.

AI was used to support brainstorming, dashboard planning, DAX measure ideas, data modelling questions, and visual build order. For example, I used it to explore possible dashboard layouts, review DAX logic, and compare modelling options such as whether date attributes should be created in the outbreak table or handled through a separate Calendar table.

I did not apply AI suggestions automatically. I reviewed the recommendations, tested them in Power BI, and validated decisions against the dataset documentation and the goals of the analysis.

Using AI helped me move faster, but the final modelling decisions, dashboard structure, and interpretation of results were reviewed and validated as part of the analysis process.

Future Improvements

This dashboard could be extended in several ways.

One future improvement would be to refresh the analysis with later 2026 data, especially July 2026 or later. This would make it possible to check whether outbreak patterns continued to decrease and whether later months showed any change after major public events.

Another improvement would be to add a dynamic subtitle that updates based on the selected Month Year filter. This would make it clearer which time period is reflected in the KPI cards and visuals.

Additional dashboard pages could also be created for deeper analysis, such as:

- a historical trend page comparing multiple years
- a COVID-19 specific trend page
- a causative agent analysis page
- an institution-level monitoring page
- a duration analysis page focused on long-running outbreaks

The dashboard could also be expanded with geographic analysis if address-level data is cleaned and prepared for mapping. This could help identify whether outbreaks are concentrated in specific areas of Toronto.

These improvements would make the project stronger as an analytical dashboard and could support future versions of the portfolio project.

End of case study